Fiscal Year:	FY 2018	**Task Last Updated:**	FY 09/25/2018
PI Name:	Davis, Catherine M. Ph.D.		
Project Title:	Neurobehavioral and CNS-Related Physiological Changes Following Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Henry M. Jackson Foundation		
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Comments:	Campus address (Jan 2022): Department of Pharmacology and Molecular Therapeutics, Uniformed Services University of the Health Sciences, 4301 Jones Bridge Road, Bethesda, MD 20814. NOTE: PI formerly at Johns Hopkins University; moved to Henry M. Jackson Foundation for the Advancement of Military Medicine in fall 2020.		
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-RADIATION
Start Date:	01/01/2015	**End Date:**	06/30/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Extended to 6/30/2018 per S. Monk/HRP/LaRC (Ed., 12/26/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hienz, Robert Ph.D. (Johns Hopkins University)		
Grant/Contract No.:	NNX15AC71G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Assessing the biological consequences of living in the space radiation environment represents one of the highest priority areas of NASA research. Of critical importance is the need for an assessment of the vulnerabilities of the central nervous system (CNS) leading to functional neurobehavioral changes during long-term space missions, and the development of effective countermeasures to such risks. The present proposal addresses this need via the application of an animal model to: 1) monitor indices of circadian function (i.e., spontaneous locomotor activity, body temperature) over various time points following irradiation; 2) determine the degree to which changes in circadian function are correlated with the severity of behavioral deficits in a rat psychomotor vigilance task; and 3) examine the underlying neurochemical alterations that accompany these radiation-induced changes in circadian function and neurobehavioral performance. Prior research has 1) identified rats that are sensitive to radiation-induced deficits in neurobehavioral function; 2) identified changes in the dopamine neurotransmitter systems of these radiation-sensitive rats; and 3) identified an increased inflammatory response in the brain from 1-14 days post-irradiation. The current proposal will determine how the immediate effects of irradiation impact subsequent neurobehavioral deficits by assessing physiological markers of circadian function that are sensitive to changes in immunological functions and the onset of pathophysiological states. Groups of animals with implanted biotelemetry transmitters will be trained on a rodent version of the human Psychomotor Vigilance Test (the rPVT) and exposed to radiation; fluctuations in spontaneous locomotor activity and body temperatures will be assessed immediately following exposure and tracked over time and compared to the neurobehavioral performances. Individual variations in these physiological responses following radiation will be assessed for use as possible biomarkers for radiation sensitivity resulting in neurobehavioral deficits. Likely mechanisms of damage to the CNS following radiation exposure will be examined using brain tissue, in addition to an examination of cytokine levels in blood and peripheral organ systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this work will provide knowledge regarding the effects of radiation on neurobehavioral function and circadian fluctuations in core body temperature and spontaneous locomotor activity. This information will enhance our understanding of radiation's effects on the CNS and how to mitigate these effects in astronauts on long-duration missions. Further, this information can be used to determine the effects of terrestrial radiation exposures on the CNS and to find methods to mitigate their adverse effects.
	FINAL REPORT--OCTOBER 2018: One cohort of 40 rats was trained on the rPVT, implanted with biotelemetry devices, and shipped to Brookhaven National Laboratory (BNL) in November 2015. These rats were sham exposed or exposed to whole-body protons (150 MeV/n) at 25 and 100 cGy, in addition to a third group exposed to 100 cGy head-only protons. Telemetry data was recorded at BNL prior to the radiation exposure and for several days after the exposure until the rats were shipped back to JHU for follow-up behavioral testing. Telemetry data was acquired 24-hrs per day for ~20 weeks post-proton exposure. During this time, rats also performed the rPVT five days per week in 30-min sessions. Similar to our previous studies and like head-only proton exposure, whole-body proton exposure resulted in significant rPVT deficits in a subset of rats, that included decreased accuracy and increased premature responding, lapses in attention, and response latencies (median and mean reaction times). Analysis of the spontaneous locomotor activity data shows that all rats, regardless of radiation exposure, display 24-hr periods of activity and this is most likely a function of their ability to entrain to the 12:12 hr light/dark cycle. Radiation exposure led to a decrease in spontaneous locomotor activity levels rats compared to sham controls. A second cohort of 40 rats was trained on the rPVT, implanted with biotelemetry devices, and shipped to BNL in June 2016. These rats were sham exposed or exposed to whole-body oxygen ions (16O, 1000 MeV/n) at 5 cGy or 25 cGy. Both 5 and 25 cGy-exposed rats displayed significant deficits in recognition memory at 301-month post-exposure; irradiated rats showed no preference for a novel odor on the test day, whereas sham-irradiated rats displayed a robust preference for this novel odor compared to a different odor experienced the previous day. Interestingly, these rats do not display deficits in novelty recognition, but are unable to retain the memory of a novel social odor over a 24-hr retention interval. At 6-months post-exposure, only the 25 cGy-exposed rats continued to display deficits in 24-hr recognition memory. The 5 cGy-exposed rats displayed intact recognition memory and performed similar to the sham-irradiated control group at this time point. The number of Ki67+ cells in the subventricular zone was similar among sham, 5 cGy and 25 cGy-exposed rats when assessed at 6-months post-exposure. Individual oxygen-exposed rats at both 5 and 25 cGy displayed significant increases in response time measures on the rPVT, primarily at the shorter wait times, suggesting that oxygen exposure results in slowed responding when there is little time to prepare for the onset of the stimulus. Response times at longer wait times were similar for all rats, suggesting that these radiation-induced deficits were specific to short stimulus onset times. These deficits are not related to motor impairments, given that these irradiated animals continued to make responses on the rPVT with response times similar to their baseline (pre-irradiation) responding at longer wait times, i.e., these rats were only slower at responding to short wait times, not longer wait times. Analysis of the spontaneous locomotor activity data shows that all rats, regardless of radiation exposure, display an activity rhythm with a 24-hr period. However, irradiated rats are less active across the 3 months following radiation exposure. Plasma cytokine levels were assessed in these rats, but no significant changes were found. A third cohort of 36 rats was trained on the rPVT, implanted with biotelemetry devices, and shipped to BNL in April 2017. These rats were sham exposed or exposed to whole-body oxygen ions (16O, 1000 MeV/n) at 1 and 10 cGy. Recognition memory tests at 1 month and 6 months were assessed. Only 10 cGy induced a deficit in 24-hr social recognition at 1 month post-exposure; however, both groups of irradiated rats displayed social recognition deficits when assessed at 6 months following exposure. rPVT deficits associated with slowed response times at shorter wait times were also found in both the 1 and 10 cGy groups and these deficits were in individual rats, with some irradiated rats displaying baseline rPVT performance levels. Cell proliferation was assessed in the subventricular zone at 1 and 6 months following exposure. While a significant age-related decrease in cell proliferation was found (greater at the 1 month time point than the 6 month time point), this effect was unrelated to radiation exposure. Plasma cytokine levels were assessed at both the 1 and 6 month time points. KC/GRO levels were significantly correlated with performance on

Task Progress:

the social recognition test, suggesting a possible peripheral marker of social recognition deficits.

A fourth cohort of 54 rats was trained on the rPVT and shipped to BNL in October 2017. Given our previous results with 16O ion (1000 MeV/n) exposure where doses of 5 cGy or lower induced variable deficits, and doses >10 cGy induced persistent social recognition deficits, the next study aimed to determine if these dose- and time-dependent effects were also LET (linear energy transfer)-dependent. For this study, we exposed groups of male Long-Evans rats to 16O ions at 350 MeV/n, using the 5 and 25 cGy doses from our previous study that induced time-and dose-dependent social recognition deficits. 16O ions at 350 MeV/n have a higher LET (20.9 keV/um) compared to 16O ions at 1000 MeV/n (14 keV/um), and are hypothesized to be more damaging to the central nervous system, which would be predicted to induce more severe and/or persistent neurobehavioral deficits. We predicted that if LET were unrelated to the radiation-induced social recognition deficits, we would see the same pattern of variable deficits at the 5 cGy dose (i.e., impairment at 1 month, but not 6 months following exposure). However, if LET were related to the etiology of these deficits (e.g., higher LET exposure could lead to damage that is less likely to be repaired), we predicted that persistent deficits would be apparent in the 5 cGy-exposed rats. Similar to exposure following 16O ions (1000 MeV/n), both doses induced significant deficits in social recognition measured at 1 month following exposure. In contrast to the results found with 1000 MeV/n 16O ions, 5 cGy 350 MeV/n 16O ion exposure resulted in persistent social recognition deficits, with performance significantly less than sham performance at the 6-month time point, thus suggesting LET was related to the etiology of these deficits. The pattern of social recognition deficits following 25 cGy 16O ion exposure was the same, regardless of the energy used. In a separate study, the role of the medial prefrontal cortex in social recognition was determined using chemogenetics. The medial prefrontal cortex (mPFC) is a frontal brain region important for many cognitive functions, including attention, decision-making, inhibitory control, reward-guided learning, habit-formation, social cognition, and different forms of memory. This brain region is theorized to be critically important for adaptive responses and anticipating emotional outcomes, given that it serves as a 'control board' that integrates a broad range of inputs. As such, damage to or altered activity in this region is often found in psychiatric conditions, including schizophrenia, depression, and substance abuse. Several pharmacological and lesion studies support the role of the mPFC in sustained attention performance, recognition memory, and social cognition. Using the chemogenetic approach termed DREADD (designer receptors exclusively activated by designer drugs), we selectively activated or inactivated the prelimbic/infralimbic regions of the mPFC to understand the role of these areas for our specific behavioral endpoints. In sham animals, we showed that the mPFC is important for encoding of the social odor stimulus, but not recall of this stimulus. We then applied this information to irradiated rats and showed that inhibition of the mPFC (via neuronal silencing), but not activation of the mPFC, attenuated the radiation-induced deficits in 24-hr social recognition. These data provide evidence to suggest that radiation exposure causes over activation of the mPFC that prevents encoding on the novel odor stimulus, which leads to impairment on the Recognition Test 24-hrs later. By inhibiting this region, irradiated rats are able to encode the novel odor stimulus and successfully identify the novel odor on the Recognition Test,, which indicates an attenuation of the radiation-induced deficit.

Conclusions:

1.) Whole-body proton (150 MeV/n) exposure results in deficits in sustained attention individual rats, such that some irradiated rats do not show any performance changes. This radiation exposure also decreases spontaneous locomotor activity across the post-radiation period, even though animals maintain 24-hr periodicity.

2.) Whole-body oxygen ion (16O, 1000 MeV/n) exposure results in deficits in social odor recognition memory at 1 month post-radiation for doses at or above 5 cGy. These same doses also result in deficits in sustained attention in individual rats that are primarily associated with slowed response times at short "wait times" on the rPVT, i.e., longer responses times when the organism has little time to prepare for the stimulus. Social recognition deficits were found at 6 months following 1 cGy, 10 cGy, and 25 cGy (1000 MeV/n) exposure. The lack of deficits at 5 cGy (1000 MeV/n) at the 6 month time point is LET-dependent, such that exposure to 5 cGy at 350 MeV/n resulted in deficits at this time point. No radiation-induced changes were found the number of Ki67+ cells in the subventricular zone in 1-25 cGy exposed rats compared to sham controls. Irradiated rats also displayed decreased levels of spontaneous locomotor activity in the post-radiation period.

3.) Using chemogenetics, we have shown that the mPFC is involved in encoding the social odor stimulus. Further, irradiated rats are unable to encode this stimulus, which results in deficits in 24-hr social recognition.

NOVEMBER 2017 REPORT:

One cohort of 40 rats was trained on the rPVT, implanted with biotelemetry devices, and shipped to Brookhaven National Laboratory (BNL) in November 2015. These rats were sham exposed or exposed to whole-body protons (150 MeV/n) at 25 and 100 cGy, in addition to a third group exposed to 100 cGy head-only protons. Telemetry data was recorded at BNL prior to the radiation exposure and for several days after the exposure until the rats were shipped back to Johns Hopkins University (JHU) for follow-up behavioral testing. Telemetry data was acquired 24-hrs per day for ~20 weeks post-proton exposure. During this time, rats also performed the rPVT five days per week in 30-min sessions. Similar to our previous studies and like head-only proton exposure, whole-body proton exposure resulted in significant rPVT deficts in a subset of rats, that included decreased accuracy and increased premature responding, lapses in attention, and response latencies (median and mean reaction times). Analysis of the spontaneous locomotor activity data shows that all rats, regardless of radiation exposure, display 24-hr periods of activity and this is most likely a function of their ability to entrain to the 12:12 hr light/dark cycle. Radiation exposure led to a decrease in spontaneous locomotor activity levels rats compared to sham controls.

A second cohort of 40 rats was trained on the rPVT, implanted with biotelemetry devices, and shipped to BNL in June 2016. These rats were sham exposed or exposed to whole-body oxygen ions (16O, 1000 MeV/n) at 5 cGy or 25 cGy. Both 5 and 25 cGy-exposed rats displayed significant deficits in recognition memory at 301-month post-exposure; irradiated rats showed no preference for a novel odor on the test day, whereas sham-irradiated rats displayed a robust preference for this novel odor compared to a different odor experienced the previous day. Interestingly, these rats do not display deficits in novelty recognition, but are unable to retain the memory of a novel social odor over a 24-hr retention interval. At 6-months post-exposure, only the 25 cGy-exposed rats continued to display deficits in 24-hr recognition memory. The 5 cGy-exposed rats displayed intact recognition memory and performed similar to the sham-irradiated control group at this time point. The number of Ki67+ cells in the subventricular zone was similar among sham, 5 cGy and 25 cGy-exposed rats when assessed at 6-months post-exposure. Individual oxygen-exposed rats at both 5 and 25 cGy displayed significant increases in response time measures on the rPVT, primarily at the shorter wait times, suggesting that oxygen exposure results in slowed responding when there is little time to prepare for the onset of the

stimulus. Response times at longer wait times were similar for all rats, suggesting that these radiation-induced deficits were specific to short stimulus onset times. These deficits are not related to motor impairments, given that these irradiated animals continued to make responses on the rPVT with response times similar to their baseline (pre-irradiation) responding at longer wait times, i.e., these rats were only slower at responding to short wait times, not longer wait times. Analysis of the spontaneous locomotor activity data shows that all rats, regardless of radiation exposure, display an activity rhythm with a 24-hr period. However, irradiated rats are less active across the 3 months following radiation exposure.

A third cohort of 36 rats was trained on the rPVT, implanted with biotelemetry devices, and shipped to BNL in April 2017. These rats were sham exposed or exposed to whole-body oxygen ions (16O, 1000 MeV/n) at 1 and 10 cGy. Recognition memory tests at 1 month and 6 months were assessed. Only 10 cGy induced a deficit in 24-hr recognition memory at 1 month post-exposure. rPVT deficits associated with slowed response times at shorter wait times were also found in both the 1 and 10 cGy groups and these deficits were in individual rats, with some irradiated rats displaying baseline rPVT performance levels.

A fourth cohort of 54 rats was trained on the rPVT and shipped to BNL in October 2017. Data collection is currently ongoing for these subjects.

Conclusions:

1.) Whole-body proton (150 MeV/n) exposure results in deficits in sustained attention individual rats, such that some irradiated rats do not show any performance changes. This radiation exposure also decreases spontaneous locomotor activity across the post-radiation period, even though animals maintain 24-hr periodicity.

2.) Whole-body oxygen ion (16O, 1000 MeV/n) exposure results in deficits in social odor recognition memory at 1 month post-radiation for doses at or above 5 cGy. These same doses also result in deficits in sustained attention in individual rats that are primarily associated with slowed response times at short "wait times" on the rPVT, i.e., longer responses times when the organism has little time to prepare for the stimulus. Only 25 cGy exposure, thus far, has resulted in memory deficits at 6-months post-exposure. No radiation-induced changes were found the number of Ki67+ cells in the subventricular zone in 5 and 25 cGy exposed rats compared to sham controls. Rats exposed to 5 and 25 cGy also showed decreased levels of spontaneous locomotor activity in the post-radiation period.

3.) A pilot study using a chemogenetic technique is being used to investigate the mechanisms of these radiation-induced deficits.

Bibliography Type:	Description: (Last Updated: 06/30/2026)
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